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CAMPUS

Amrita School of Computing

Coding Practice

Activity Sheets

(For the Batch 2023-2027)

Name:

Roll No.:

Department:

Instructions

The **Mock Test syllabus** for CSE CT | CYS | AIE | RAI program is enclosed in the attached document for your reference.

Materials for the Mock Test [Must-DO Coding Questions] is provided in a link.

It is mandatory to upload the **completed Activity sheets** before the deadline in the MS-**Form link** shared by your class Advisor.

Activity Sheet: 06 - Coding Questions from Digit Manipulation – Part 1 [Easy + Medium Questions]

Each Activity Sheets contains total of **20 Questions**.

Platform: Any Online Compiler of your choice – **GDB Compiler**

Language: C | C++ | JAVA | PYTHON

The Activity Sheet will have the coding Questions and the Test Case. Once executed in Online Compiler, take **screen shot of the code and the output** and paste it in the space provided in the activity sheet.

The Final Activity Sheet should be in **PDF Format** and the file name should be your **roll number**.

Course Instructors: CARE TEAM

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1. An odometer has 5 slots, where each slot contains a wheel. Initially, the odometer shows the number 00000. A person wants to set a number N in the odometer. He can rotate the wheels only in clockwise direction from right to left in the odometer. The value of N is passed as the input to the program. The program must print the combinations he gets when setting up the number N in the odometer.

Boundary Condition(s):

$1 \leq N \leq 99999$

Input Format:

The first line contains N.

Output Format:

The lines, each contains a 5-digit number(with leading zeroes) representing the odometer.

Example :

Input:

30151

Output:

00000

00001

00011

00021

00031

00041

00051

00151

10151

20151

30151

Code:

Output:

2. There are N water tanks arranged in a row. The maximum capacity of each water tank is M. The initial amount of water in each tank is passed as the input to the program. If it rains over the N tanks for an hour, 1 litre of water will be filled in each tank. Whenever a tank is full, a boy drains K litres of water from that the tank. The program must print the amount of water in each tank after T hours of rain as the output. The values of K and T are also passed as the input.

Boundary Condition(s):

$1 \leq N \leq 100$

$1 \leq K \leq M \leq 100$

$0 \leq \text{Amount of water in each tank} < M$

$1 \leq T \leq 1000$

Input Format:

The first line contains N and M separated by a space. The second line contains N integers separated by a space. The third line contains K and T separated by a space.

Output Format:

The first line contains N integers separated by a space.

Example :

Input:

4 30

5 15 10 25

4 11

Output:

16 26 21 28

Code:

Output

3. Positive Even Integers

The program must accept N integers and print the first K positive even integers as the output.

Note: At least K positive even integers are always present in the given N integers.

0 is neither positive nor negative.

Boundary Condition(s): $1 \leq K \leq N \leq 100$ - $10^5 \leq$ Each integer value $\leq 10^5$

Input Format: The first line contains N. The second line contains N integers separated by a space. The third line contains K.

Output Format: The first line contains the first K positive even integers separated by a space.

Example:

Input:

10

-84 36 23 8 -25 0 14 -78 42 20

4

Output:

36 8 14 42

Explanation: Here $N = 10$ and $K = 4$.

The given 10 integers are -84, 36, 23, 8, -25, 0, 14, -78, 42 and 20.

The first 4 positive even integers are 36, 8, 14 and 42.

Hence the output is 36 8 14 42

Code:

Output

4. Product - First Digit & Last Digit

The program must accept N non-zero integers as the input. The program must modify the given integers based on the following conditions. - If an integer is odd, the program must multiply the integer with its unit digit. - If an integer is even, the program must multiply the integer with its first digit. Finally, the program must print the sum of the N modified integers as the output.

Boundary Condition(s):

$1 \leq N \leq 100$

$-10^6 \leq \text{Each integer value} \leq 10^6$

Input Format:

The first line contains N. The second line contains N integers separated by a space.

Output Format: The first line contains the sum of the N modified integers.

Example:

Input: 5

15 260 -46 117 -9

Output:

1149

Code:

Output

5. The program must accept N integers and print the adjacent integers whose difference is less than or equal to X or more than Y. If there are no such adjacent integers, the program must print -1.

Boundary Condition(s): $2 \leq N \leq 100$ $-10000 \leq \text{Each integer value} \leq 10000$ $1 \leq X, Y \leq 10000$

Input Format: The first line contains N. The second line contains N integer values separated by a space. The third line contains X and Y separated by a space.

Output Format: The first line contains -1 or the lines, each contains the adjacent integers whose difference is less than or equal to X or more than Y.

Example:

Input :

6

-5 10 55 1000 1100 1300

20 200

Output:

-5:10

55:1000

Code:

Output

6. Remove Minimum - Matrix

The program must accept an integer matrix of size $R \times C$ containing unique integers as the input. The program must remove the minimum integer in each row of the matrix. Then the program must remove the minimum integer in each column of the matrix. Finally, the program must print the modified matrix as the output.

Boundary Condition(s):

$3 \leq R, C \leq 50$

$1 \leq \text{Matrix element value} \leq 10^4$

Input Format: The first line contains R and C separated by a space. The next R lines, each contains C integers separated by a space.

Output Format: The first $R-1$ lines, each contains $C-1$ integers separated by a space.

Example :

Input: 4 4

28 11 18 17

14 20 29 26

25 15 22 13

27 19 23 30

Output:

28 18 26

25 29 22

27 23 30

Code:

Output

7. Last Integer - Eight Directions

The program must accept an integer matrix of size $R \times C$ and an integer X as the input. If the integer X is present inside the border of the matrix, the program must print the last integers in all 8 directions of X (north, north-east, east, south-east, south, south-west, west and north-west) as the output. Else the program must print -1 as the output. If the integer X occurs two or more times inside the border of the matrix, the program must consider the first occurring integer.

Boundary Condition(s):

$3 \leq R, C \leq 50$

$1 \leq \text{Matrix element value}, X \leq 1000$

Input Format: The first line contains R and C separated by a space. The next R lines, each containing C integers separated by a space. The $(R+2)$ nd line contains X .

Output Format: The first line contains the last integers in all 8 directions of X .

Example:

Input:

```
5 6
47 48 63 37 29 72
78 60 67 13 46 30
65 72 95 55 52 76
86 62 88 48 80 91
31 49 81 76 27 95
55
```

Output:

```
37 72 76 95 76 49 65 48
```

Code:

Output

8. Prime Numbers - Subsets

The program must accept an integer N as the input. The program must print the prime numbers which are formed using the subsets of N in the order of their occurrence as the output. If there is no prime number, the program must print -1 as the output.

Boundary Condition(s): $2 \leq N \leq 10^8$

Input Format: The first line contains N.

Output Format: The first line contains the prime numbers which are formed using the subsets of N or -1.

Example:

Input:

5423

Output:

5 2 23 3

Explanation: All the subsets of 5423 are given below.

5

54

542

5423

4

42

423

2

23

3

There 4 prime numbers present in the above subsets. Hence the output is 5 2 23 3

Code:

Output

9. Tenth Digit Greater

The program must accept N integer values and print the values whose tenth digit is greater than the previous value. (All the values are greater than 9). As the first value does not have a previous value, consider the last value as the previous value for the first value.

Boundary Condition(s): $2 \leq N \leq 100$ $10 \leq \text{Each integer value} \leq 10^5$

Input Format: The first line contains N. The second line contains N integers separated by a space.

Output Format: The first line contains the integer value(s) separated by a space.

Example:

Input:

7

20 72 204 727 1073 153 207

Output: 20 72 727 1073

Explanation: The given 7 integers are 20, 72, 204, 727, 1073, 153 and 207.

The first integer 20 does not have a previous value, so the last integer 207 is considered as the previous value.

The tenth digit of 20 is greater than the tenth digit of 207. So 20 is printed.

The tenth digit of 72 is greater than the tenth digit of 20. So 72 is printed.

The tenth digit of 204 is not greater than the tenth digit of 72.

The tenth digit of 727 is greater than the tenth digit of 204. So 727 is printed.

The tenth digit of 1073 is greater than the tenth digit of 727. So 1073 is printed.

The tenth digit of 153 is not greater than the tenth digit of 1073.

The tenth digit of 207 is not greater than the tenth digit of 153.

Hence the output is 20 72 727 1073

Code:

Output

10. Matrix Diagonal Print Bottom Left

The program must accept a matrix of size $R \times C$ and print them diagonally starting from bottom left.

Boundary Condition(s):

$1 \leq R, C \leq 100$

Input Format: The first line contains R and C separated by a space. The next R lines contain C integer values each separated by a space.

Output Format: $R+C-1$ lines containing integer values separated by a space.

Example:

Input:

```
3 4
10 20 30 40
50 51 52 53
60 70 80 90
```

Output:

```
60
50 70
10 51 80
20 52 90
30 53
40
```

Code:

Output

11. Non-overlapping Submatrices Sum

The program must accept an integer matrix of size $R \times C$ and two integer X , Y as the input. The program must print the sum of integers in all $X \times Y$ non-overlapping submatrices as the output.

Boundary Condition(s):

$2 \leq X \leq R \leq 50$

$2 \leq Y \leq C \leq 50$

$1 \leq \text{Matrix element value} \leq 1000$

Input Format:

The first line contains R and C separated by a space. The next R lines, each contains C integers separated by a space. The $(R+2)$ th line contains X and Y separated by a space.

Output Format: The first line contains the integer values representing the sum of integers in all the $X \times Y$ non-overlapping submatrices.

Example:

Input:

10 9

5 7 5 3 2 9 5 1 6

3 7 6 2 5 9 6 9 2

3 7 8 9 7 3 9 6 6

6 2 3 9 1 4 5 2 9

9 4 3 7 8 5 2 2 5

9 1 8 9 6 3 2 9 2

8 2 3 8 9 7 3 9 1

1 8 6 9 4 6 5 3 1

3 7 8 9 7 9 2 6 9

8 9 2 5 9 5 3 1 8

4 4

Output: 85 83 95 83

Code:

Output

12. Consecutive 1s between 0s

The program must accept two integers X and Y as the input. The program must find the binary representation of the integers X and Y. Then the program must find the longest common part having only 1s between two 0s. Then the program must replace all 1s with 0s except all the non-overlapping occurrences of the longest common part from left to right in the binary representations of X and Y. If there is no such common part, then the program must not change the values of X and Y. Finally, the program must print the sum of the modified values of X and Y as the output.

Boundary Condition(s): $1 \leq X, Y < 2^{31}$

Input Format: The first line contains X and Y separated by a space.

Output Format: The first line contains the sum of the modified values of X and Y.

Example:

Input: 2934 735

Output: 966

Code:

Output

13. Piggy Bank & Toy

A boy wants to save money to buy a toy. He puts money in his piggy bank every day. Every day, he puts 1 rupee more than the previous day. Every sunday, he starts with a new value 1 rupee more than the previous sunday. The cost of the toy is T rupees. The boy already has X rupees in his piggy bank. On the first sunday, he puts Y rupees in his piggy bank. The values of T, X and Y are passed as the input to the program. The program must print the number of days he needs to buy the toy as the output.

Boundary Condition(s):

$1 \leq T \leq 10^4$

$0 \leq X < T$

$1 \leq Y \leq 100$

Input Format: The first line contains T, X and Y separated by a space.

Output Format: The first line contains the number of days he needs to buy the toy.

Example:

Input: 170 50 10

Output: 10

Code:

Output

14. Serial Killer - N Houses

There are N houses in a row. A serial killer plans to kill people in these houses. If he targets Xth house, he will kill 2 persons in that house and 1 person from nearby by houses (left and right). If there is no person in the Xth house, he will not kill any person in the nearby houses. The program must accept N integers representing the number of persons in the N houses and M integers representing the positions of the houses he is targeting. The program must print the number of persons remaining in the N houses after the series of murders as the output.

Boundary Condition(s):

$1 \leq N \leq 100$

$0 \leq \text{Number of persons in each house} \leq 1000$

$1 \leq M \leq 1000$

Input Format: The first line contains N. The second line contains N integers separated by a space. The third line contains M. The fourth line contains M integers separated by a space.

Output Format: The first line contains the number of persons remaining in the N houses after the series of murders.

Example:

Input:

5

12 6 8 2 4

3

2 5 4

Output: 11 4 6 0 1

Code:

Output

15. Integers - Parity Sort

The program must accept N integers as the input. The program must sort the integers based on the parity (Even parity followed by Odd parity). If two or more integers having the same parity, then the program must sort those integers in ascending order. If the number of 1s in the binary representation of an integer is even, then it represents Even Parity. If the number of 1s in the binary representation of an integer is odd, then it represents Odd Parity.

Boundary Condition(s):

$2 \leq N \leq 100$

$1 \leq \text{Each integer value} \leq 1000$

Input Format: The first line contains N. The second line contains N integers separated by a space.

Output Format: The first line contains the sorted integers based on the given conditions.

Example:

Input:

6

15 25 387 440 48 80

Output:

15 48 80 387 25 440

Code:

Output

16. Trees - Circular Ground

There are N trees planted circularly in a ground. Every day, the gardener will cut down X trees circularly so that the height of each tree will be reduced by 1. Similarly, each tree will grow, thus increasing the height by 1 each day (The trees cut down by the gardener will not grow on that day). If the height of a tree is 1, then he cannot cut down the tree even if it is present in the X trees that he wants to cut down. The program must accept N integers representing the height of N trees and the value of X as the input. The program must print the height of the N trees after T days. The value of T is also passed as the input to the program.

Boundary Condition(s):

$1 \leq X \leq N \leq 100$

$1 \leq T \leq 100$

Input Format: The first line contains N. The second line contains N integers separated by a space. The third line contains X and T separated by a space.

Output Format: The first line contains the height of the N trees after T days separated by a space.

Example:

Input:

5

10 7 5 8 6

3 4

Output:

8 5 5 8 6

Code:

Output

17. Rotate Digits - 180 Degrees

The program must accept an integer N as the input. The program must form an integer X by rotating the each digit 180 degrees based on the following conditions. - The digits 2 and 5 rotate to each other. - The digits 6 and 9 rotate to each other. - For the remaining digits (0, 1, 3, 4, 7 and 8), they remain the same. Finally, the program must print the sum of N and X as the output.

Boundary Condition(s):

$1 \leq N \leq 10^8$

Input Format: The first line contains N.

Output Format: The first line contains the sum of N and X.

Example:

Input: 10672

Output: 21647

Code:

Output

18. Cow & Goat Fodder

A farmer has X kilograms of wheat and Y kilograms of corn. He wants to prepare the fodder for cows and goats. The fodder for a cow is made using 2 kilograms of wheat and 1 kilogram of corn. The fodder for a goat is made using 1 kilogram of wheat and 2 kilograms of corn. The profit from selling each of the fodder is 1 rupee. The program must accept the values of X and Y as the input. The program must print the maximum profit the farmer can earn as the output.

Boundary Condition(s):

$0 \leq X, Y \leq 10^8$

Input Format: The first line contains X and Y separated by a space.

Output Format: The first line contains the maximum profit the farmer can earn by selling the fodders.

Example:

Input: 4 4

Output: 2

Code:

Output

19. Matrix Spiral – Alternate Direction Removal

The program must accept an integer matrix of size $R \times C$. You must traverse the matrix in spiral order, but with a twist:

- Traverse normally in spiral (right $\rightarrow$ down $\rightarrow$ left $\rightarrow$ up)
- After completing each layer, reverse the direction of traversal for the next layer
- Print all elements in the order visited

Example:

Input:

```
4 4
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

Output:

```
1 2 3 4 8 12 16 15 14 13 9 5 6 7 11 10
```

Code:

Output

20. Binary Matrix – Largest Rectangle of 1s

The program must accept a binary matrix of size $R \times C$ (containing only 0s and 1s).

The program must print the area of the largest rectangle containing only 1s.

Input Format:

- First line: R and C
- Next R lines: C binary values (0 or 1)

Output Format:

- Print a single integer → maximum rectangular area of 1s

Input:

```
4 5
1 0 1 1 1
1 1 1 1 1
0 1 1 1 0
1 1 1 1 0
```

Output:

```
9
```

Code:

Output